Quiz 1 - NLP

19L-2448
Hamza Yousuf

Question 1) Training Corpus:

<s> I am Sam </s>
<s> Sam I am </s>
<s> I am Sam </s>
<s> I do not like green eggs and Sam </s>

$\boxed{N = 21}$

(a) Test Sentence:

<s> I am like Sam </s>

(i) Unigram (without Smoothing)
(ii) Bigram (" ")
(iii) Trigram (" ")
(iv) Bigram (Laplace Smoothing)
(v) Trigram (" ")
(vi) Bigram Language Model with Linear interpolation
$\lambda_1 = 0.7$ (Bigram weight)
$\lambda_2 = 0.3$ (Unigram weight)

(b) Calculate Perplexity of test sentence using bigram model with Laplace smoothing.

Question 2) A dictionary can be used to identify spelling errors. How does language modeling help in task of spelling correction.

Q1 (a) (i) Unigram Modeling

$$P(\langle s\rangle I \text{ am like Sam } \langle /s\rangle) = P(I).P(am).P(like).P(Sam)P(\langle /s\rangle)$$

$$= \frac{4}{21} \cdot \frac{3}{21} \cdot \frac{1}{21} \cdot \frac{4}{21} \cdot \frac{4}{21}$$

$$= \frac{64}{1361367} \approx 0.0000047$$

Bigram Modeling

$P(\text{I like Sam } \langle/s\rangle) = P(I|\langle s\rangle)\cdot P(am|I)\cdot P(like|am)\cdot P(sam|like)\cdot P(\langle/s\rangle|sam)$.

$$= \frac{3}{4}\cdot\frac{3}{4}\cdot\frac{0}{3}\cdot\frac{0}{1}\cdot\frac{3}{4} = \boxed{0}$$

Trigram Modeling.

$P(\langle s\rangle I \text{ am like Sam } \langle/s\rangle) = P(I|\langle s\rangle\langle s\rangle)\cdot P(am|\langle s\rangle I)\cdot P(like|I\,am)\cdot P(sam|am\ like)\cdot P(\langle/s\rangle|like\ Sam)$

$$\frac{3}{4}\cdot\frac{2}{3}\cdot\frac{0}{2}\cdot\frac{0}{0}\cdot\frac{0}{0} = \boxed{0}$$

(iv) Bigram (Smoothed). $|V| = 9$

$\hat{P}(\text{Test Sentence}) = \hat{P}(I|\langle s\rangle)\cdot\hat{P}(am|I)\cdot\hat{P}(like|am)\cdot\hat{P}(sam|like)\cdot\hat{P}(\langle/s\rangle|sam)$

$$= \frac{3+1}{4+9}\cdot\frac{3+1}{4+9}\cdot\frac{0+1}{3+9}\cdot\frac{0+1}{1+9}\cdot\frac{3+1}{4+9}$$

$$= \frac{4}{13}\cdot\frac{4}{13}\cdot\frac{1}{12}\cdot\frac{1}{10}\cdot\frac{4}{13} = \frac{4^3}{13^3\cdot 12\cdot 10} = \frac{64}{263640}$$

$$\approx \boxed{0.000242}$$

(v) Trigram (Smoothed)

$\hat{P}(\text{Test Sentence}) = \hat{P}(I|\langle s\rangle\langle s\rangle)\hat{P}(am|\langle s\rangle I)\cdot\hat{P}(like|I\,am)\cdot\hat{P}(sam|am\ like)\cdot P(\langle/s\rangle|like\ Sam)$

$$= \frac{3+1}{4+9}\cdot\frac{2+1}{3+9}\cdot\frac{0+1}{2+9}\cdot\frac{0+1}{0+9}\cdot\frac{0+1}{0+9}$$

$$= \frac{4}{13}\cdot\frac{3}{12}\cdot\frac{1}{11}\cdot\frac{1}{9}\cdot\frac{1}{9} = \frac{12}{138996} \approx \boxed{0.000086}$$

(vi) Bigram Linear Interpolation.

$\hat{P}(\text{Test}) = \hat{P}(I|\langle s\rangle)\hat{P}(am|I)\hat{P}(like|am)\hat{P}(sam|like)\hat{P}(\langle/s\rangle|sam)$

$$= \left[0.7\left(\tfrac{3}{4}\right)+0.3\left(\tfrac{4}{21}\right)\right]\cdot\left[0.7\left(\tfrac{3}{4}\right)+0.3\left(\tfrac{8}{21}\right)\right]\cdot\left[0.7\left(\tfrac{0}{3}\right)+0.3\left(\tfrac{1}{21}\right)\right]\cdot\left[0.7\left(\tfrac{0}{1}\right)+0.3\left(\tfrac{4}{21}\right)\right]\cdot\left[0.7\left(\tfrac{3}{4}\right)+0.3\left(\tfrac{4}{21}\right)\right]$$

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$$= [0.525 + 0.057]\left[\frac{21}{4} + \frac{0.9}{21}\right]\left[\frac{0.3}{21}\right]\left[\frac{1.2}{24}\right]$$

$$= [0.525 + 0.057][0.525 + 0.0428][0.014][0.057][0.525 + 0.0$$

$$= 0.582 \times 0.095 \times 0.014 \times 0.057 \times 0.582 = \boxed{0.0000}$$

$$PP(\text{Sentence}) = P(\text{Sentence})^{-\frac{1}{N}}$$

$$= (0.000242)^{-\frac{1}{5}}$$

$$= \boxed{5.2873}$$

Q2 There are two scenarios. If the word with wrong spelling does not make a word then its probability will be zero from the Language model and we can replace with the word having max probability. Second scenario is where the word is an actual word but not the intended one hence Language model will help in getting the rightword by giving max probability to the right word according to the context.